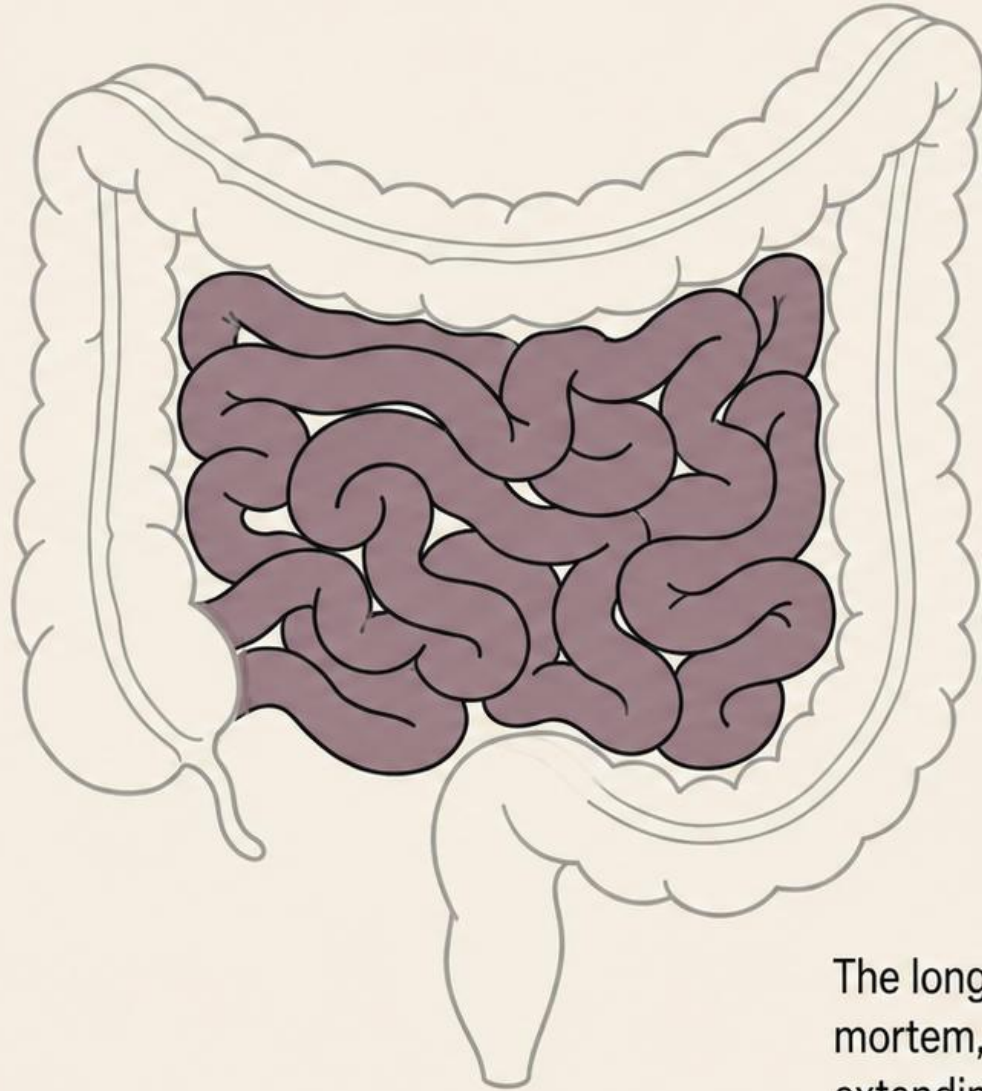


The Small Intestine: Anatomy, Ecosystem and ROP Approach

Chapter 14 — The systemic, mechanical, neurological and immune crossroads.



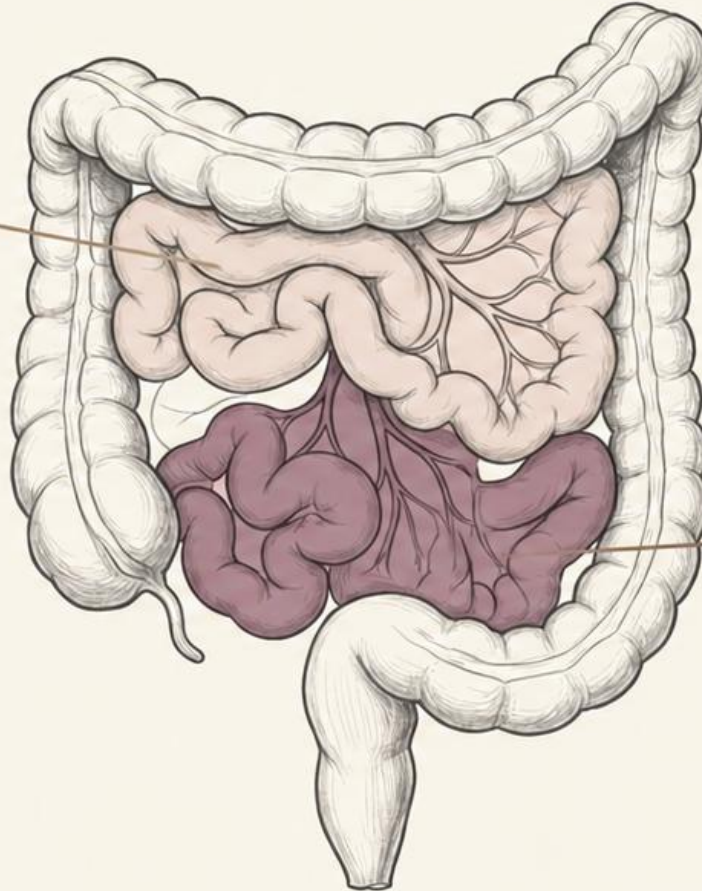
The longest organ in the body (6 to 7 meters post-mortem, shortened in vivo by muscle tone), extending from the pylorus to the ileocecal valve.

Topography: The Small Intestine within the Colonic Frame

The **jejunum-ileum** (6 to 7 meters in the cadaver) comprises 15 to 16 intestinal loops distributed into two distinct functional groups.

Jejunum

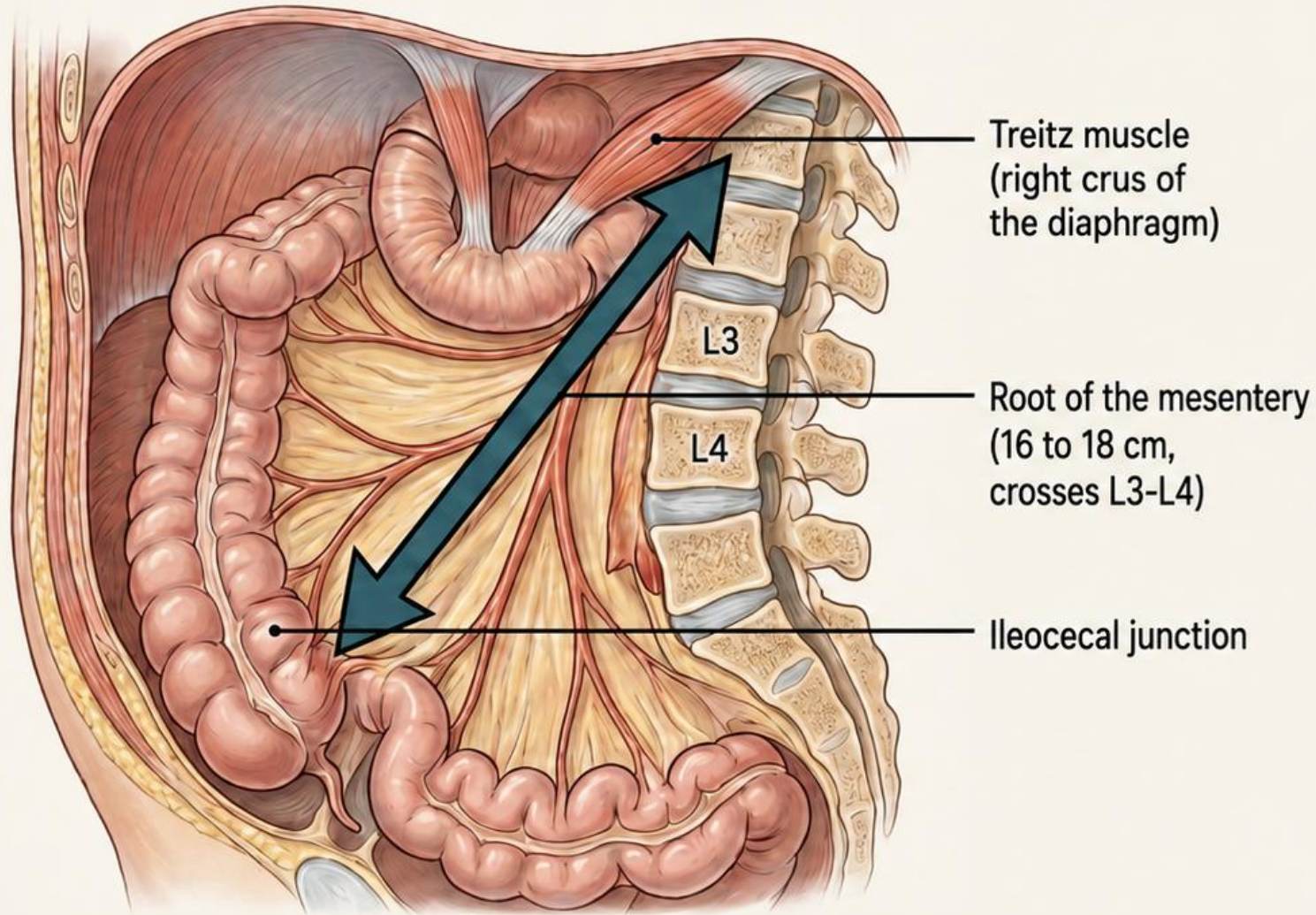
- **Location:** Left upper quadrant.
- **Orientation:** Loops and vascular system oriented horizontally.
- **Relations:** Overlies the descending colon.
- **Characteristics:** Typically the more developed group.



Ileum

- **Location:** Right lower quadrant.
- **Orientation:** Loops and vascular system oriented vertically.
- **Relations:** Leaves the ascending colon free.

The mesentery and its suspensory mechanism

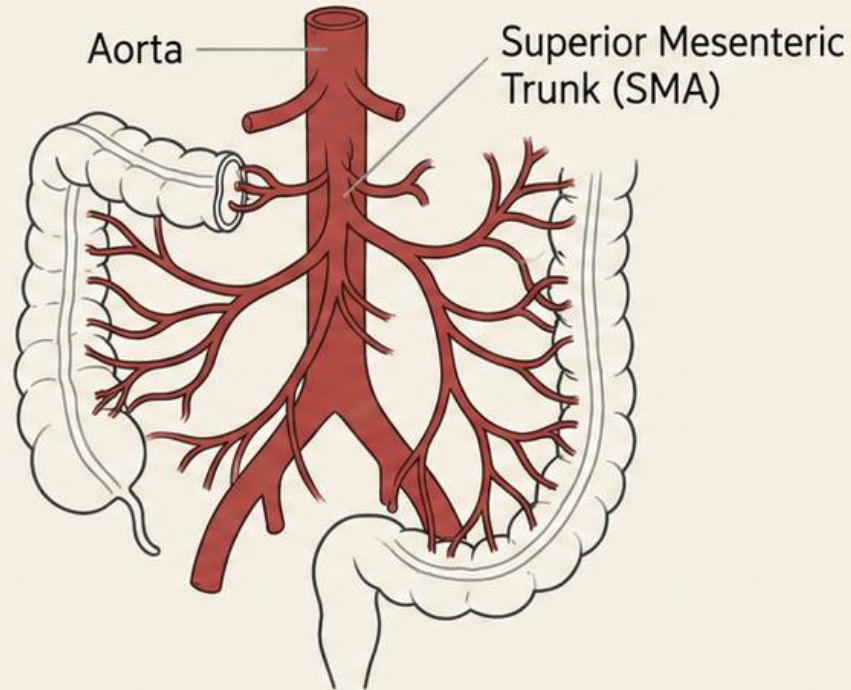


The mesentery is a double-layered fold of the peritoneum connecting the small intestine to the posterior abdominal wall; it contains the blood vessels, lymphatic vessels, and nerves supplying the intestine.

Relevance in ROP (Nutcracker Syndrome)

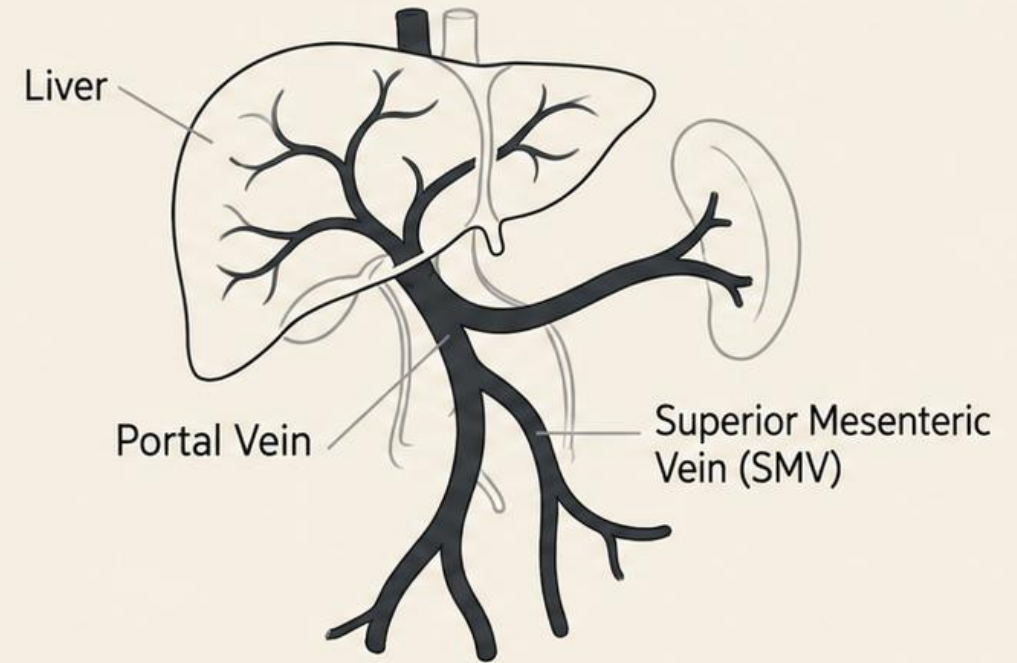
- A **decrease** in the longitudinal tension of the mesenteric root may promote aortomesenteric **compression**: compression of the fourth portion of the duodenum and/or of the left renal vein between the aorta and the superior mesenteric artery
- **Clinical consequence**: delayed emptying of duodenal contents and gastro-duodenal reflux.

The axis of irrigation and drainage: Superior Mesenteric Trunk



Arterial Supply (Superior mesenteric artery)

- Origin: T12–L1 disc (below the celiac trunk)
- Supplies the small intestine, the ascending colon, and the proximal 2/3 of the transverse colon



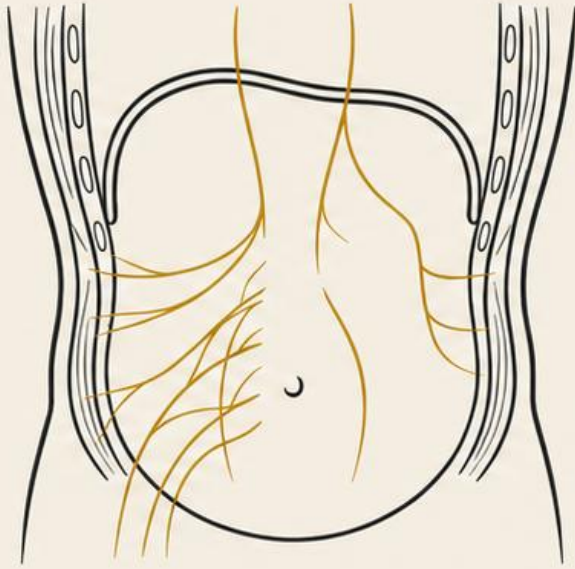
Venous Drainage (Portal System)

- Carries venous blood rich in absorbed nutrients directly to the liver for detoxification and metabolism.

Practical Note: The pulse of the SMA can be felt at the level of the 3rd part of the duodenum, to the right of the umbilicus.

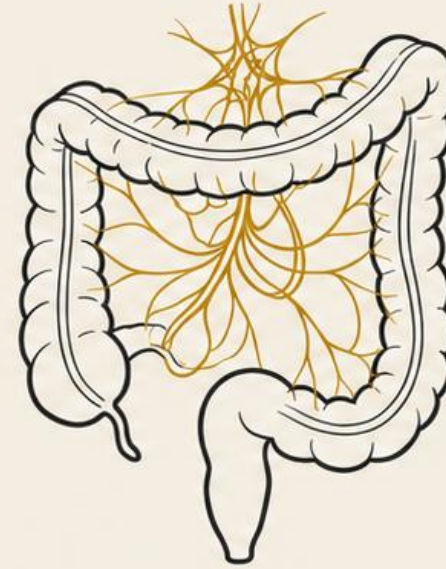
The double neurological reality of the peritoneum

Parietal Peritoneum (Somatic Innervation)



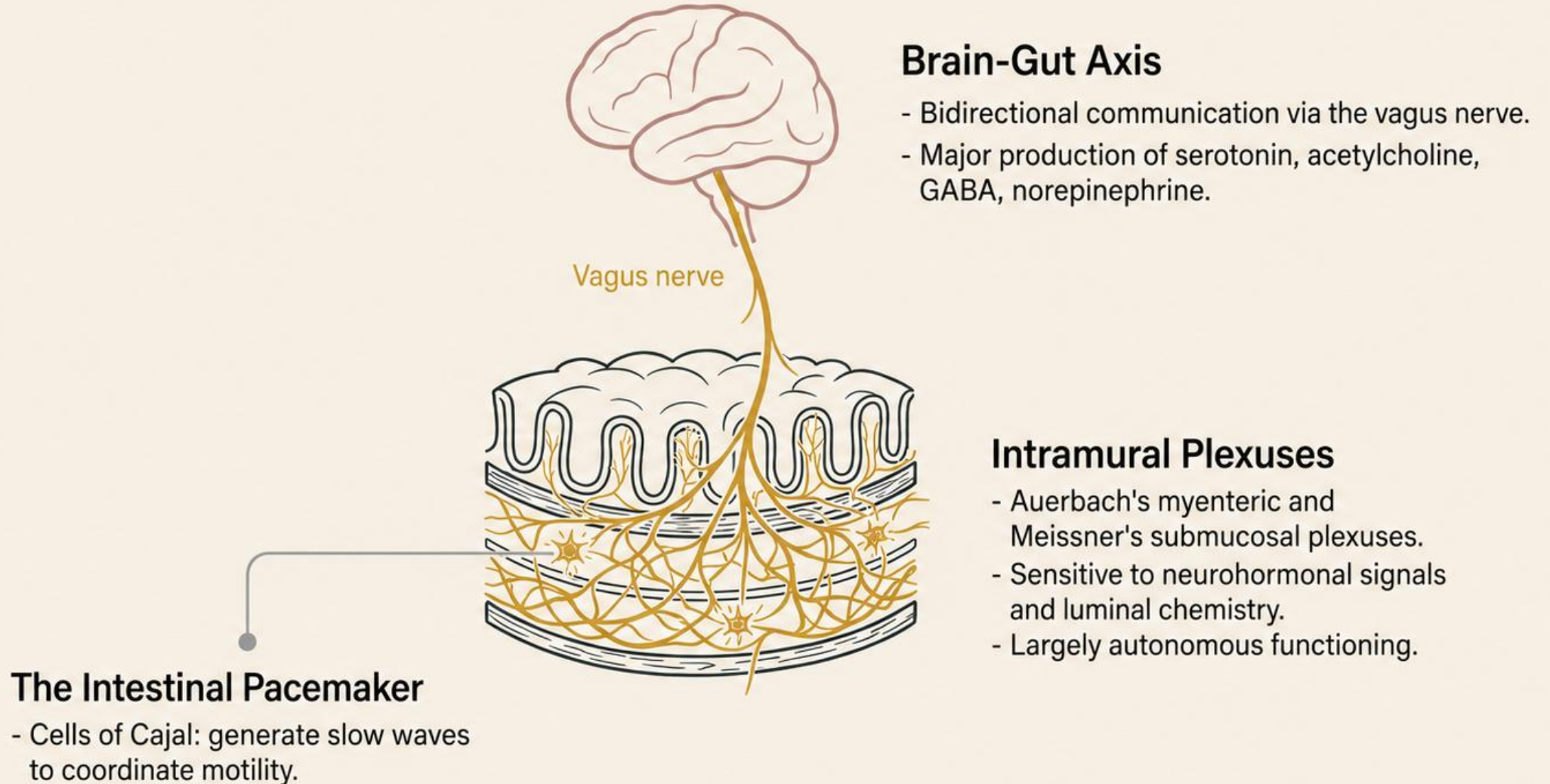
- Pathways: Phrenic nerves, last 6 intercostal nerves, lumbar plexus.
- Sensitivity: Sharp (pain, pressure, temperature).
- ROP implication: Projection of peritoneal pain toward the osteo-musculo-articular system (cervical, lumbar, scapular).

Visceral Peritoneum (Autonomic Innervation)



- Pathways: Sympathetic system (T8-T11, splanchnic nerves) and parasympathetic system (vagus nerve).
- Sensitivity: Low for touch/incision. Stimulated by stretching (distension) and chemical stimuli.
- ROP implication: The sympathetic system inhibits peristalsis (stress); the vagus promotes it.

The Second Brain: Autonomy of the Enteric Nervous System (ENS)



The "Second Brain": Enteric Nervous System

Intramural Plexuses

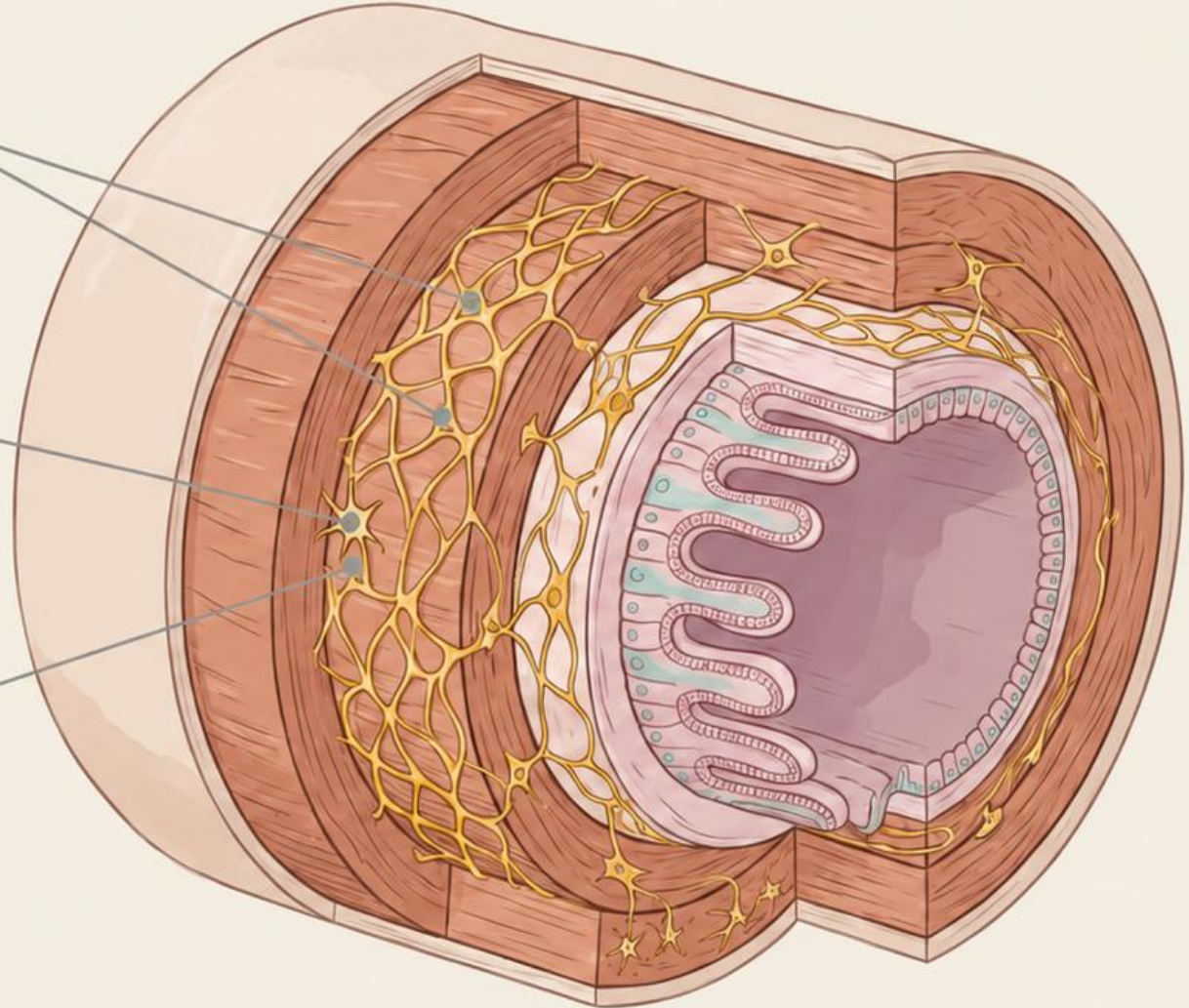
Auerbach's myenteric and Meissner's submucosal plexuses. Provide near-autonomous regulation of digestive functions.

Cajal Cells

The intestinal pacemaker. Generate slow waves that coordinate motility independently of chyme-induced peristalsis.

Neurochemical Production

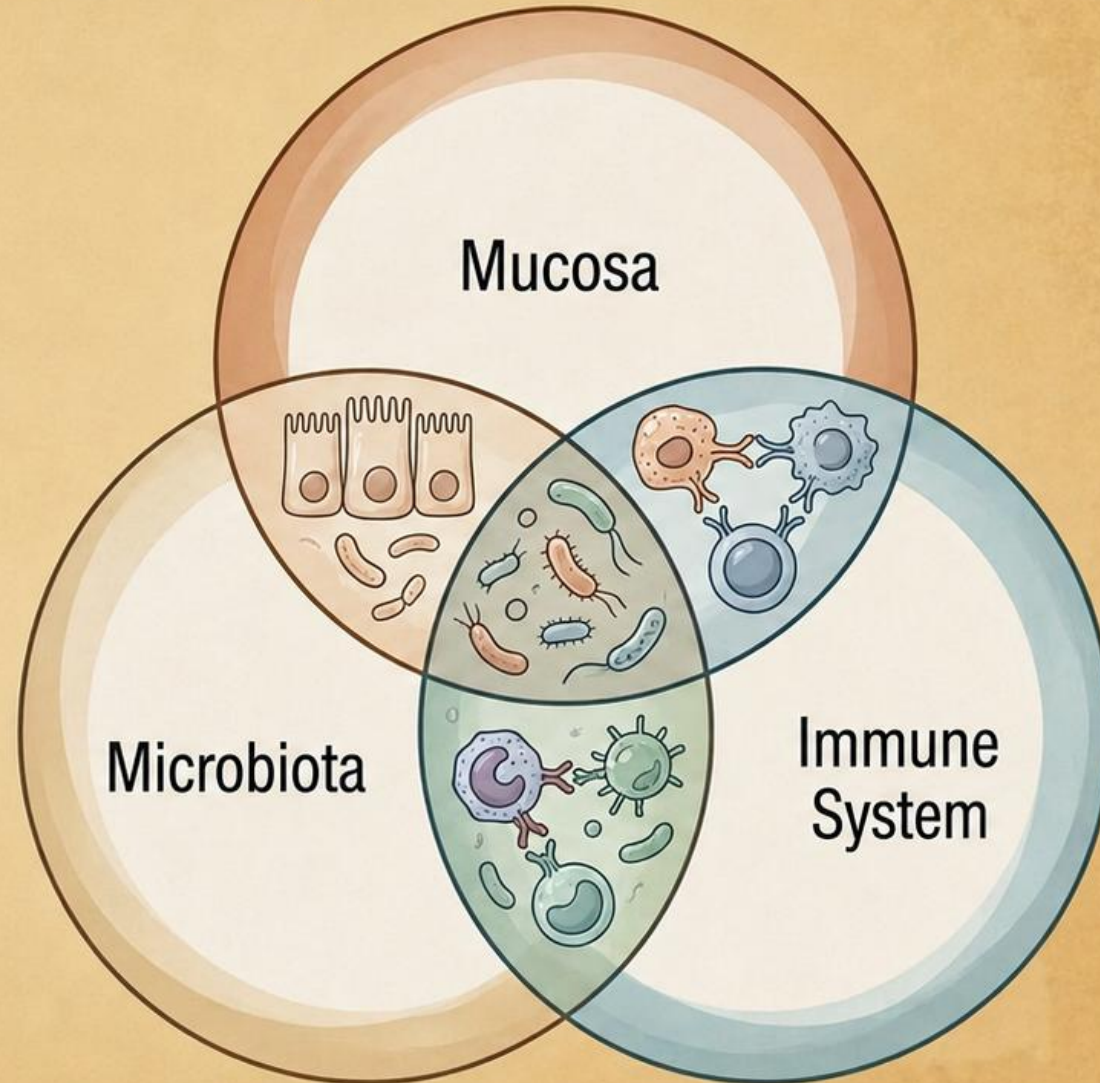
Major synthesis of serotonin (mood regulation shared with the brain), acetylcholine, norepinephrine, and GABA.



The Intestinal Ecosystem: Symbiosis and Immunity

The Microbiota (Flora):

- Approximately 38 billion bacteria.
- A unique signature for each individual, established from birth.
- A commensal/symbiotic relationship essential to digestion and immunity.



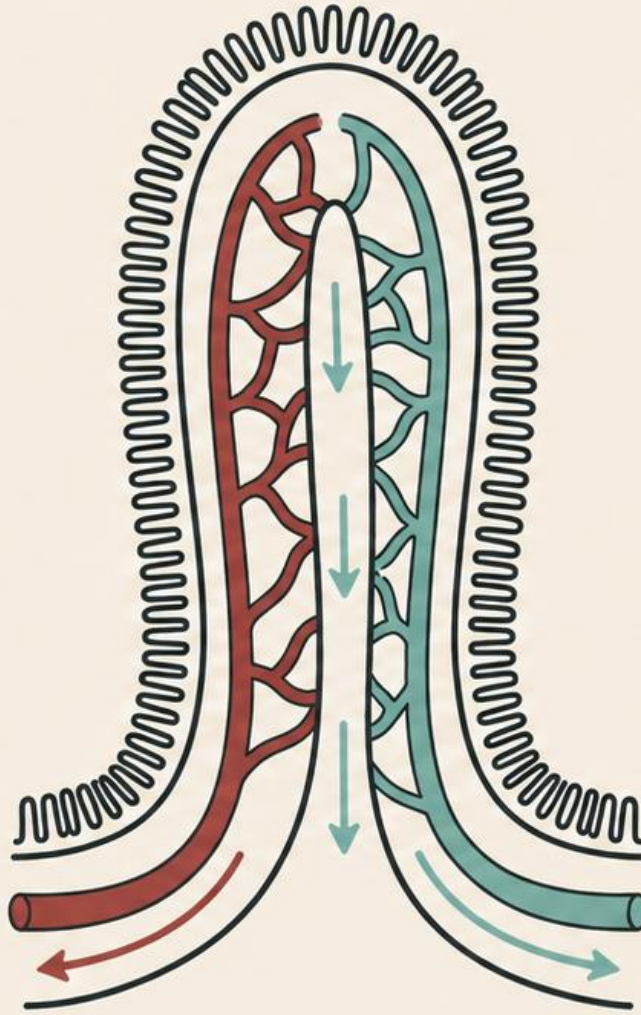
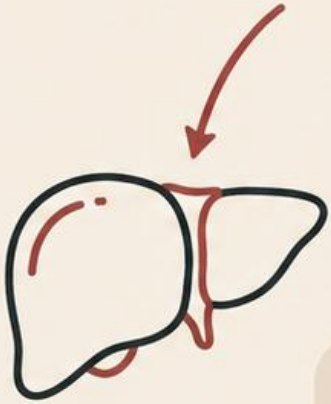
Immunity (GALT):

- Gut-Associated Lymphoid Tissue: Lines the mucosa (includes Peyer's patches).
- Mast cells & Paneth cells: Release antimicrobial peptides.
- Microbial load maintained well below that of the colon under healthy conditions.

The Dual Pathway of Micronutrient Absorption

Blood Pathway (Capillaries)

- Absorbed: Water, simple sugars, amino acids, water-soluble vitamins.
- Destination: Transported directly to the liver via the portal system.



Lymphatic Pathway (Lacteals)

- Absorbed: Fats, long-chain proteins, fat-soluble vitamins (A, D, E, K).
- Destination: Lymphatic network -> Cisterna chyli -> Thoracic duct.



Relevance in ROP: A foot reflex action on the liver and the lymphatic system qualitatively supports absorption and immunity.

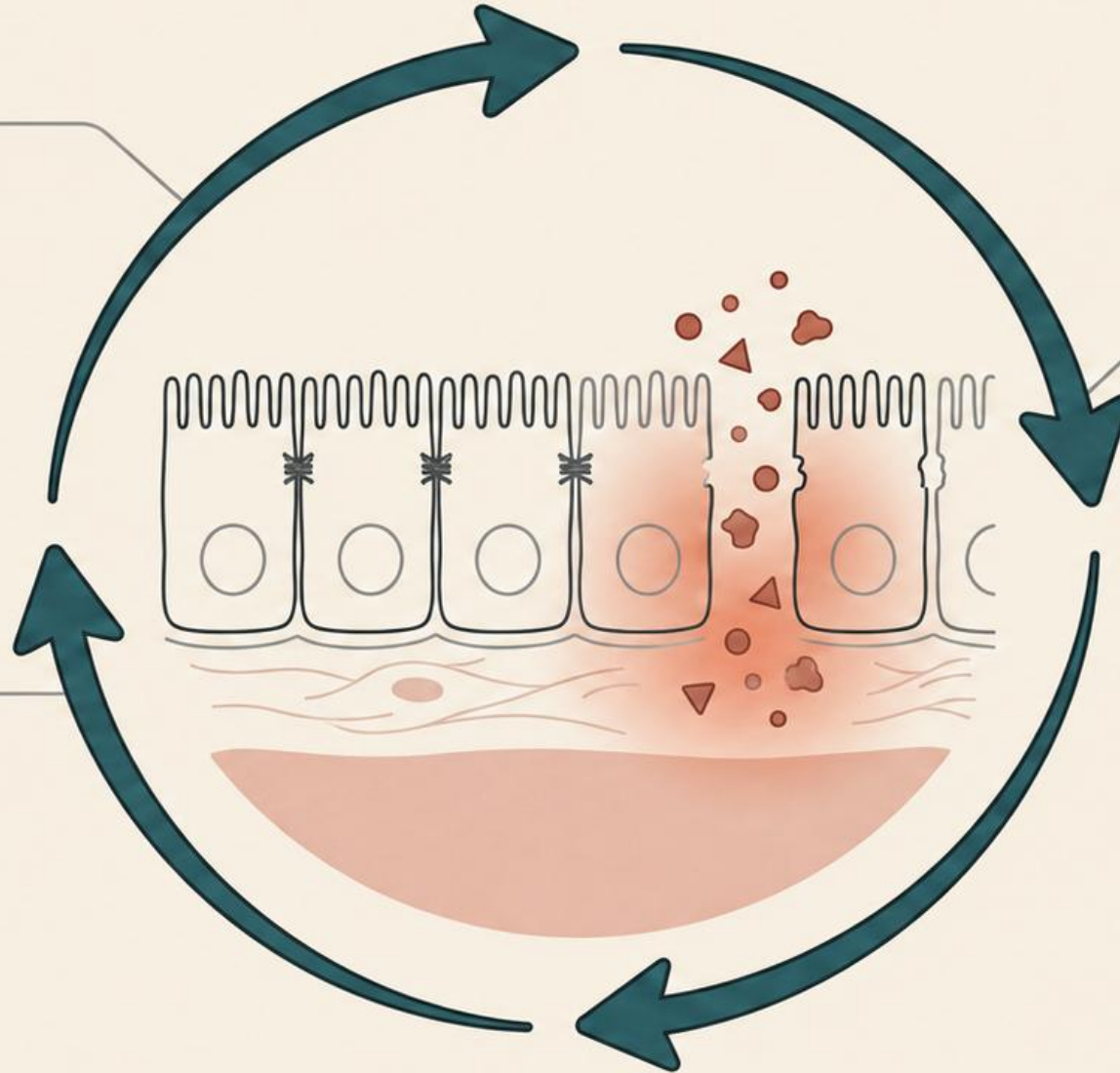
The Spiral: Dysbiosis and Hyperpermeability

Triggers

Chronic stress
(sympathetic
vasoconstriction),
ultra-processed foods,
NSAIDs, antibiotics.

Hyperpermeability

Alteration of tight junctions.
Passage of toxins and
incompletely degraded
proteins, triggering
inflammation.

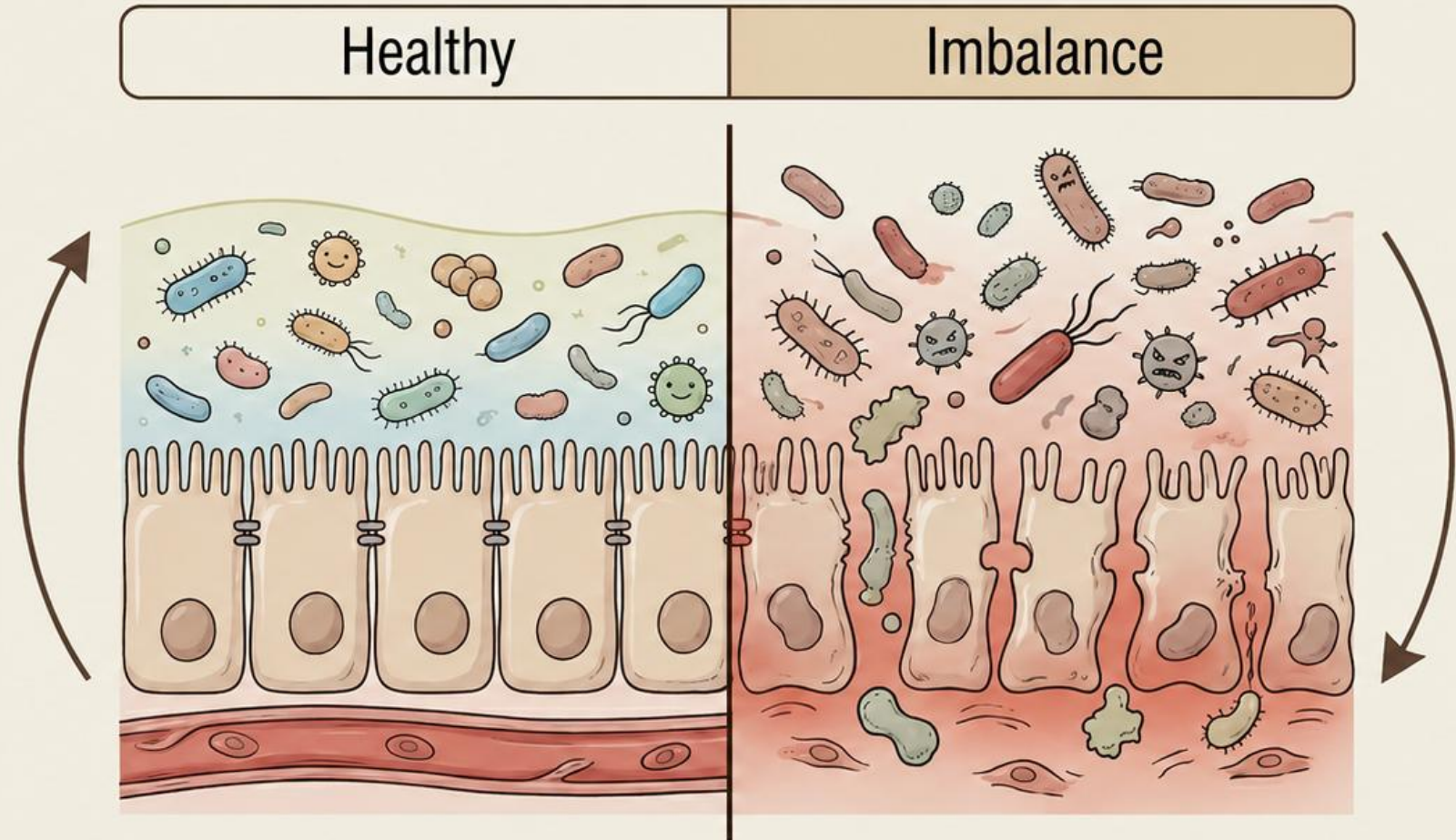


Systemic Consequences

Hepatic overload
(Seignalet's clogging
theory). The saturated
liver redirects waste
to the tissues, causing
osteo-articular pain,
tendon pain, and
allergies.

Imbalance: Hyperpermeability and Dysbiosis

The Vicious Cycle: Disruption of tight junctions (Leaky Gut) and alteration of the microbiota (Dysbiosis) perpetuate each other, shifting from physiological fermentation to abnormal putrefaction.



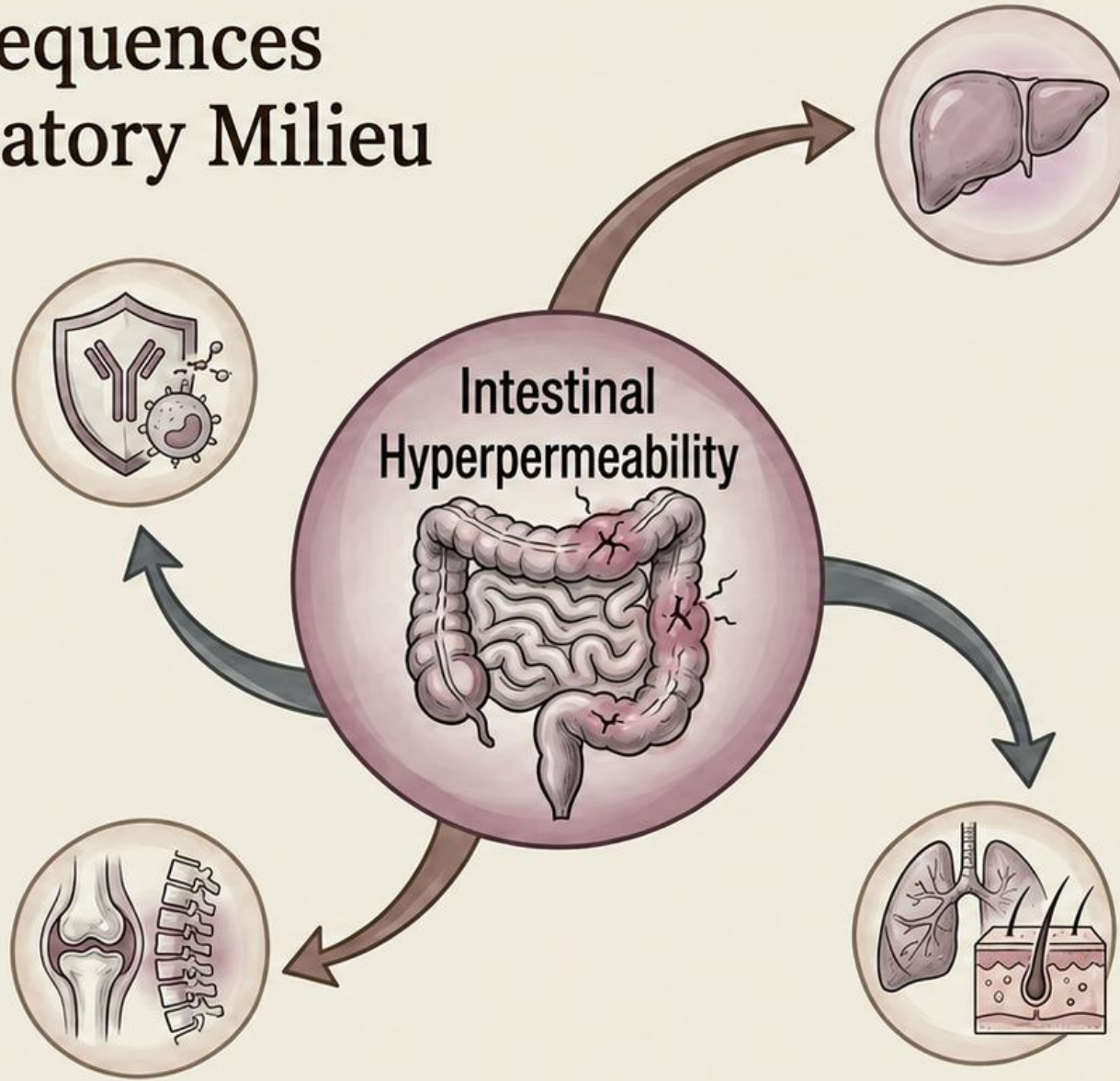
Triggering Factors:

- Chronic stress: Sympathetic activation altering blood flow.
- Medications: NSAIDs, antibiotics, corticosteroids.
- Lifestyle: Ultra-processed diet, sugars, extreme endurance exercise (splanchnic blood flow steal).
- Biomechanics (ROP): Vertebral fixations, fibrosis of peritoneal attachments.

Systemic Consequences of the Inflammatory Milieu

Immune Dysregulation:
Abnormal exposure of the immune system to food antigens -> allergies, hypersensitivity (histamine).

Musculoskeletal Pain:
Low back pain, arthralgia, fibromyalgia (theory of tissue congestion).



Hepatic Overload:
Increased portal passage of toxins increases the liver's detoxification workload.

Associated Conditions:
Asthma, sinusitis, cystitis, autoimmune diseases (associated factor).

ROP Perspective: In the presence of joint pain without evident trauma, investigate visceral involvement.

Warning Signs and Diagnosis of Exclusion

Immediate Medical Referral Is Imperative in the Presence of:

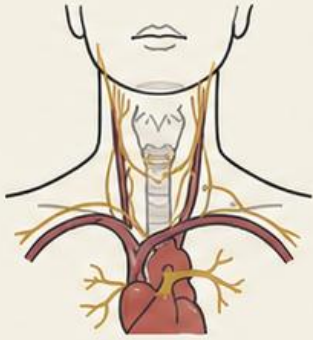
- - Fever, red or black blood in the stools.
- - Significant and unexplained weight loss.
- - Troisier's node (left jugulo-subclavian junction): oncologic suspicion.
- - Ice-cube sign: suggestive of ascites.
- - Irreducible strangulated inguinal hernia.
- - Obstruction / Ileus: Cessation of stool and gas passage. May be mechanical (tumor, adhesion) or paralytic (atony due to peritoneal irritation).

ROP Treatment Protocol: Order of Management

Address the terrain before the local symptom.



**Step 1: MRP
& Brain**
(Diencephalon, Brain
Trunk).



Step 2:
Autonomic Nervous
System (ANS)
(Vagal Nerve
Balance).



**Step 3: Liver
& Drainage**
(Support Metabolism
and Excretory
Function).



**Step 4: Intestine
Small & Large**
(Motility, Immune Aspects,
Mucosal Integrity).



Step 5:
Osteo-Articular
(Pelvis, Sacrum, Ilium,
Spine).

Objective

Support the body's self-regulatory capacities without replacing allopathic medical follow-up.
Progression is gradual and reassessed.

ROP Intervention Summary: Small Intestine Protocol

Step 1: Listening and Induction (Limbic System)

Action: One thumb on the Small Intestine area, the other on the Limbic Brain area.

Objective: Soothe the viscerο-emotional load (stress, anxiety, need for safety) and reactivate vagal tone.



Step 2: Structural Release (Viscero-Somatic Relationships)

Action: Treatment of vertebral and costal fixations.

Target zones: T10-T12 junction (sympathetic) and L1-L2 (lumbar plexus, root of the mesentery).



Step 3: Locoregional Treatment (Visceral)

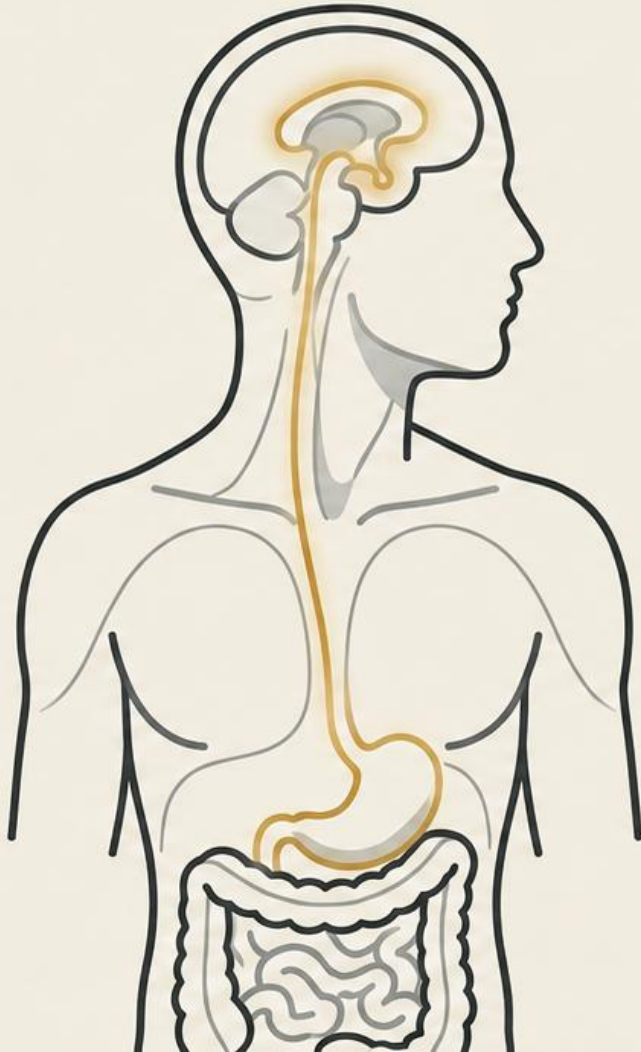
Action: Release of suspension tensions and stimulation of peristalsis.

Target zones: Diaphragm (Treitz), Stomach (reflex), Root of the mesentery (longitudinal tension), and Jejunum/Ileum mapping.

The ROP approach restores mobility, innervation, and the ecosystem: a global response to the maxim "All disease begins in the intestine".

Viscero-Emotional Profile: The 'Intestin Person'

The mucous membranes are seen in ROP as emotion receivers-transmitters.



Physical Signs

- Morning fatigue
- Low back pain / knee pain
- Bowel transit disorders
- Hypersensitivity to osteo-muscular fixations

Emotional Traits

- Need for visceral security and fixed references.
- Overprotective toward the senses, meticulous.
- Anxiety often masked by a need for reassurance or control.
- Changeable mood, easily contradicted.